

Route-F P54: Exact Invariant Attribution, Nonuniform Scalar Control and the Radial Fermion Kernel

P54PQ-v2 four-dimensional mainline

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Abstract

We resolve the scalar one-loop radial mass and kinetic corrections into the actual invariants of the unchanged P54PQ-v2 action, retaining all interference. Multihomogeneity determines the full radial Hessian jets from one ambient Hessian per invariant, avoiding a repeated numerical derivative scan. Fixed-propagator attribution is distinguished from constrained parameter changes, which require new spectra, counterterms and light-doublet tuning. This calculation concerns engineering control of candidate backgrounds, not a fit to observed particle masses. The radial Majorana determinant can be evaluated independently of a complete flavor fit, but a full gauged physical stability verdict still requires the consistently regulated momentum-dependent bosonic kernel.

1 Unchanged theory and scope

The action, 328 canonical real scalar coordinates and tensor normalizations are exactly those of `verify_p54_p1_hessian_spectrum.py`. The reference point is

$$r_0 = (w, \sigma, v_s) = (1, 0.1265, 0.25)\omega, \quad g_U = \mu_0/\omega = 0.5626028899853225. \quad (1)$$

Here “original point” means this broken background, not the symmetric configuration $X = 0$. We use the same $\text{DR}/\overline{\text{MS}}$, background-field Landau, fixed-VEV invariant-mass counterterm prescription as P-REN1 and P-IR1. Let $X = Rr$, with

$$G = R^T R = \text{diag}(12/5, 2, 1), \quad H = V_0''(X), \quad T_j = \partial_{r_j} H, \quad U_{jk} = \partial_{r_j} \partial_{r_k} H. \quad (2)$$

Thus a coordinate $\sigma\sigma$ entry must be divided by two to obtain its value on a unit canonical σ direction. The 38 tree soft modes are 33 gauge Goldstones, one physical PQ mode and the four real components of one tuned complex light doublet. There are 290 positive scalar modes. Their mass weights are recomputed for every candidate, not copied from the original point.

2 One Hessian per invariant: exact block Euler reconstruction

Write the real-parameter slice of the action as

$$V_0 = \sum_{\alpha} p_{\alpha} I_{\alpha}, \quad (\Phi, \Sigma, H_{10}, S) = (w\Phi_0, \sigma\Sigma_0, 0, v_s S_0). \quad (3)$$

The real and imaginary canonical coordinates of each complex field belong to the same scaling sector. If I_{α} has multidegree $d_{\alpha} = (d_{\Phi}, d_{\Sigma}, d_H, d_S)$, the Hessian block with external fields in sectors A, B has degrees

$$e_{\alpha, AB, k} = d_{\alpha, k} - \delta_{Ak} - \delta_{Bk}. \quad (4)$$

Negative degrees signify a zero derivative, not a negative-power continuation. At $H_{10} = 0$, blocks with remaining $e_H > 0$ also vanish. For the other blocks and positive radial coordinates,

$$H_{AB}^\alpha(r') = H_{AB}^\alpha(r_0) \prod_{k \in \{\Phi, \Sigma, S\}} (r'_k/r_{0k})^{e_k}, \quad (5)$$

$$(T_j^\alpha)_{AB} = \frac{e_j}{r_j} H_{AB}^\alpha, \quad (U_{jk}^\alpha)_{AB} = \frac{e_j(e_k - \delta_{jk})}{r_j r_k} H_{AB}^\alpha. \quad (6)$$

These identities follow by differentiating each homogeneous polynomial block. They are exact for the fixed angular background, not a truncated Taylor approximation or a fit. At a zero radius one must use polynomial coefficients directly instead of dividing by that radius.

Couplings	Multidegree (Φ, Σ, H, S)
$\mu^2; c; a, b$	$(2, 0, 0, 0); (3, 0, 0, 0); (4, 0, 0, 0)$
$\nu^2; \lambda_0, \lambda_2, \lambda_4, \lambda'_4$	$(0, 2, 0, 0); (0, 4, 0, 0)$
$\alpha, \beta; \xi_0^2; \xi_1, \xi_2$	$(2, 2, 0, 0); (0, 0, 2, 0); (0, 0, 4, 0)$
$\xi_3; \gamma_1, \gamma_2, \eta_3$	$(1, 0, 2, 0); (0, 2, 2, 0)$
$\eta_0, \eta_2; \eta_1$	$(2, 0, 2, 0); (0, 3, 1, 0)$
$\mu_s^2; \chi_1; \chi_2$	$(0, 0, 0, 2); (0, 0, 0, 4); (0, 2, 0, 2)$
$\chi_3; \chi_4; \chi_5$	$(2, 0, 0, 2); (1, 2, 0, 1); (0, 0, 2, 2)$
$\chi_6; \chi_7$	$(0, 0, 2, 1); (1, 0, 2, 1)$

For a complex coupling, $cI + c^*I^* = 2c_R\Re I - 2c_I\Im I$. This checkpoint varies the real directions only. A CP-phase study requires the independent imaginary basis and new angular-stationarity tests. Also, a vanishing radial potential does not imply a vanishing ambient Hessian: transverse states can couple strongly to that radius. Conversely, ξ_1, ξ_2 multiply H_{10}^4 and have identically zero H, T, U on this entire radial slice. Changing them cannot repair this particular one-loop radial failure, although they affect boundedness, nonzero- H_{10} backgrounds and other amplitudes.

Proposition 1 (A structural blind spot of radial tests). *Passing every radial mass/kinetic test on $H_{10} = 0$ does not imply boundedness of the scalar action.*

Proof. Set $\Phi = \Sigma = S = 0$. The quartic potential is

$$V_4 = \xi_1(H^\dagger H)^2 + \xi_2|H^T H|^2. \quad (7)$$

For a complex null vector $H = t(1, i, 0, \dots)/\sqrt{2}$, one has $H^T H = 0$ and $V_4 = \xi_1 t^4$. Thus $\xi_1 < 0$ makes the potential unbounded below without changing any of the radial H, T, U computed above. A real H similarly requires $\xi_1 + \xi_2 \geq 0$. Since $0 \leq |H^T H|^2/(H^\dagger H)^2 \leq 1$, these two conditions characterize nonnegativity of this pure- H quartic. They are only necessary for the full action; equality still requires lower-degree terms on flat rays, and mixed-field directions have not been bounded. $\square$

Both necessary inequalities are now explicit rejection gates. The global mixed-quartic and competing-vacuum problems remain separate; additional radial decimal accuracy cannot replace either one.

3 Attribution with all interference retained

Diagonalize only the complete action:

$$Q^T H Q = \text{diag}(m_i^2), \quad \tilde{T}_j^\alpha = Q^T T_j^\alpha Q, \quad \tilde{U}_{jk}^\alpha = Q^T U_{jk}^\alpha Q. \quad (8)$$

Here component jets include the actual coefficient p_α . Individual invariants need not have positive Hessians and do not define separate physical determinants. Fix Q, m_i^2 and the hard projector only for the following attribution. Define

$$A_i = m_i^2 [\ln(m_i^2/\mu_0^2) - 1] \quad (i \text{ hard}), \quad A_i = 0 \quad (i \text{ soft}), \quad (9)$$

$$L(a, b; s) = \int_0^1 \ln \frac{(1-x)a + xb + x(1-x)s}{\mu_0^2} dx, \quad s = p_E^2, \quad (10)$$

$$W(a, b) = \int_0^1 \frac{x(1-x)}{(1-x)a + xb} dx. \quad (11)$$

In the local hard kernels below, at least one internal line is hard. Hard-soft terms are retained. Pure soft-soft terms instead contain $L(0, 0; s) = \ln(s/\mu_0^2) - 2$ and are not a local kinetic coefficient.

The scalar tadpole and common finite counterterm for component α are linear:

$$t_j^\alpha = \frac{1}{32\pi^2} \sum_i A_i (\tilde{T}_j^\alpha)_{ii}, \quad (12)$$

$$\mathcal{A}(r) = -\text{diag}(12w/5, \sigma, v_s), \quad \Delta p^\alpha = -\mathcal{A}(r)^{-1} t^\alpha, \quad (13)$$

$$R^T H_{\text{CT}}^\alpha R = -\text{diag}(t_w^\alpha/w, t_\sigma^\alpha/\sigma, t_v^\alpha/v_s). \quad (14)$$

Hence define

$$C_{jk}^\alpha = \frac{1}{32\pi^2} \sum_i A_i (\tilde{U}_{jk}^\alpha)_{ii} + (R^T H_{\text{CT}}^\alpha R)_{jk}. \quad (15)$$

The ordered bubble and kinetic interference blocks are

$$B_{jk}^{\alpha\beta} = \frac{1}{32\pi^2} \sum'_{il} (\tilde{T}_j^\alpha)_{il} (\tilde{T}_k^\beta)_{li} L(m_i^2, m_l^2; 0), \quad (16)$$

$$Z_{jk}^{\alpha\beta} = \frac{1}{32\pi^2} \sum'_{il} (\tilde{T}_j^\alpha)_{il} (\tilde{T}_k^\beta)_{li} W(m_i^2, m_l^2). \quad (17)$$

The prime excludes only two-soft-line pairs. All ordered blocks are saved, including $\alpha \neq \beta$. Their exact sums give

$$H_{S+\text{CT}} = \sum_\alpha C^\alpha + \sum_{\alpha\beta} B^{\alpha\beta}, \quad \Delta Z_S = \sum_{\alpha\beta} Z^{\alpha\beta}. \quad (18)$$

The joint (α, j) kinetic Gram matrix is positive semidefinite, but a cross block need not be positive. The signed equal-share ledger

$$\mathcal{M}^\alpha = C^\alpha + \frac{1}{2} \sum_\beta (B^{\alpha\beta} + B^{\beta\alpha}), \quad (19)$$

$$\mathcal{Z}^\alpha = \frac{1}{2} \sum_\beta (Z^{\alpha\beta} + Z^{\beta\alpha}) \quad (20)$$

is a specified accounting convention, not a separately measurable “contribution percentage.” Its sums reconstruct the full matrices.

3.1 Why the ledger is not a parameter derivative

If κ_α scales only vertices at fixed propagators,

$$F(\kappa) = \sum_\alpha \kappa_\alpha C^\alpha + \sum_{\alpha\beta} \kappa_\alpha \kappa_\beta B^{\alpha\beta}, \quad \partial_{\kappa_\alpha} F|_1 = C^\alpha + \sum_\beta (B^{\alpha\beta} + B^{\beta\alpha}). \quad (21)$$

Even this frozen derivative differs from the signed allocation. Actual parameter changes also differentiate masses, eigenprojectors, loop weights and the stationarity/light-doublet constraints. The three dependent mass terms illustrate the distinction: their cubic/quartic jets vanish, but they still change all propagators.

4 Actual invariant ledger at the original point

The exact reconstruction, random independently reweighted vertex tests, full joint kinetic Gram positivity and source comparisons pass 445/445 checks. The first run required 27 unit-coefficient Hessians (the two H^4 zeros are algebraic) and one new independent full-action radial point. Subsequent runs use those caches without new evaluations. For the canonical σ direction $e_\sigma/\sqrt{2}$ and the G -normalized leading kinetic eigenvector z , the signed ledger is

Invariant	$\mathcal{M}_{\sigma\sigma}/(2\omega^2)$	$z^T \mathcal{Z} z$	$z^T Z^{\alpha\alpha} z$
λ'_4	-0.29460220	0.89223875	0.37939853
λ_2	-0.25234309	0.85449316	0.29839039
λ_4	-0.19391061	0.70984670	0.21519366
λ_0	-0.02977324	0.09307175	0.00598139
α	-0.00000790	0.06652577	0.00226620
β	0.00000601	-0.01376498	0.00206069

The three transverse Σ quartics account for 2.45657860 of the leading 2.61031989 kinetic insertion (94.11% in this specified allocation), and $-0.74085590\omega^2$ of the scalar-plus-CT canonical σ curvature (96.13%). These percentages are not invariant observables. In particular all kinetic self terms sum to only 0.90706619; discarding interference misses 1.70325369.

The radial restriction of the actual tensor action contains $\lambda_0\sigma^4/4$, but not these three transverse quartics. At a stationary background

$$(H_0)_{\sigma\sigma} = 2\lambda_0\sigma^2, \quad R^T H^{\lambda_2, \lambda_4, \lambda'_4} R = 0. \quad (22)$$

The latter projected numerical norms are below 2×10^{-16} at the original card. Thus transverse quartics can be weakened without simultaneously weakening this tree radial restoring term. This is a directional reason to test nonuniform changes, not a guarantee of success: the propagator spectrum still changes.

5 Constrained nonuniform re-evaluation

At each separately declared candidate card, assemble the full polynomial H, T, U by Eq. (6). Re-solve the three mass parameters by

$$\delta p = -\mathcal{A}(r)^{-1} \nabla_r V_0, \quad H \mapsto H + H_{\text{mass}}(\delta p), \quad (23)$$

and tune ξ_0^2 to the first physical eigenvalue crossing, after removing the 34 symmetry directions. Verify the four extra zero modes have $(C_2^{SU(3)}, C_2^{SU(2)}, Y^2) = (0, 3/4, 1/4)$ and all other scalar tree modes are positive. Then recompute every scalar/vector mass, radial jet contraction and the common finite tadpole shift.

The engineering tests are

$$H_b^{\text{hard}} > 0, \quad \rho_H = \max |\lambda(H_b^{\text{hard}} - H_0, H_0)| < 0.3, \quad \rho_Z = \lambda_{\text{max}}(\Delta Z_S, G) < 0.3. \quad (24)$$

The margin 0.3 is declared, not a theorem bounding omitted loops. Passing these radial tests does not prove full nonradial stability, boundedness from below, a physical pole, or a valid unification fit. No observed mass or flavor data are used, and no default matching input is overwritten.

5.1 First-stage cards: weakening quartics alone is insufficient

The six declared cards use $(\lambda_2, \lambda_4, \lambda'_4) \mapsto \tau(\lambda_2, \lambda_4, \lambda'_4)$. Except as labelled, $r = r_0$ and all other independent coefficients are unchanged. L2 doubles λ_0 ; X multiplies χ_4 by 0.03; S30 also changes σ/ω to 0.3.

Card	τ	$\min \lambda(H_b^{\text{hard}}, G)/\omega^2$	ρ_H	ρ_Z
T30	0.30	-0.434370	32.3860	0.902652
T10	0.10	-0.106794	8.59012	0.713427
T03	0.03	-0.026376	2.84877	1.099954
T03 L2	0.03	-0.067918	3.31182	1.370839
T03 X	0.03	-0.026349	2.84547	1.099232
T03 X S30	0.03	-0.052148	1.66116	0.632098

All six fail. Their actual action reconstruction, necessary BFB negative controls and 80 independent bubble integrals pass 183/183 numerical checks; rejection is a physical-control result, not a failed code test. At T03 X S30, a separate evaluation of the final retuned full action, three coordinate-direction jets and a mixed-direction jet confirms that the invariant assembly is not merely an assumed input identity.

The nonmonotonic kinetic insertion has a concrete spectral origin. Between T10 and T03 the positive scalar gap drops from 0.00883111 to $0.00267622\omega^2$, while the leading kinetic eigenvector's canonical Φ fraction rises from 0.667676 to 0.824065. In this new mass basis the signed α allocation rises from 0.437312 to 1.081219 (the accompanying β interference is negative). Thus the original leading- Σ quartic problem is replaced by a large mixed-coupling response in the Φ direction. Merely shrinking a vertex does not uniformly shrink a loop with a simultaneously closing propagator gap.

For example, any positive hard squared-mass gap Δ gives

$$W(a, b) \leq \frac{1}{6\Delta} \quad (a, b \geq \Delta), \quad W(0, b) \leq \frac{1}{2\Delta} \quad (b \geq \Delta). \quad (25)$$

Therefore a useful control estimate must constrain both the vertex norm and the inverse gap. A fixed-parameter derivative of a mass is not a derivative along the reanchored stationary-mass curve; confusing the two would incorrectly remove precisely the response measured here.

5.2 Second-stage joint mixed-coupling probes

After identifying this response, two additional coarse cards were declared, not optimized: $\lambda_0 = 0.2$ as an absolute value; $(\lambda_2, \lambda_4, \lambda'_4, \chi_4)$ are multiplied by 0.01; (α, β, χ_2) by 0.1. All other independent coefficients are retained, with $w = 1$, $v_s = 0.25$ and $\sigma = 0.3$ or 0.5 . Dependent masses and the tree doublet are re-solved as above.

σ/ω	$\min \lambda(H_b^{\text{hard}}, G)/\omega^2$	ρ_H	ρ_Z
0.3	-0.00317747	1.176951	1.131599
0.5	0.02299103	0.538986	1.899032

The 36 checks pass with zero new Hessian evaluations, but neither card passes the declared control gates. The second is an explicit example of positive hard curvature without small loop insertions. Its leading kinetic vector is 99.9798% in the canonical Φ direction. Neither is retained, so no new independent full-action Hessian calculation is spent on these two rejected cards; their assembly uses the checked invariant library. This is the end of this bounded eight-card test, not the start of an unrestricted optimization.

5.3 A conditional inverse-gap obstruction, not a decimal fit

On the ray changing only the three transverse quartics at fixed r_0 , two mutually orthogonal scalar eigenspaces have numerically verified projector identities (the following numerical matrices use $\omega = 1$)

$$P_{50} H P_{50} = 12\tau\sigma^2 P_{50}, \quad P_{36} H P_{36} = \frac{58}{5}\tau\sigma^2 P_{36}, \quad (26)$$

$$P_j T_w P_j = \frac{9}{25} P_j, \quad j = 50, 36. \quad (27)$$

The projectors are pure Σ , with ranks indicated by their subscripts. Their entries admit exact integer-over-eight projector certificates. The action-to-projector identities above are checked against the actual tensor action, but a symbolic or interval certificate for those action identities is not yet supplied. The following implication is therefore an exact conditional bound, not an unconditional theorem about the entire $\text{Spin}(10)$ action.

The separately tested invariant coefficients give the useful mass branch formula

$$m_\kappa^2 = -\frac{\nu^2}{2} + (\lambda_0/2 + \tau\kappa)\sigma^2 + \frac{6}{5}(\alpha - \beta)w^2 + 60\chi_2 v_s^2, \quad (N, \kappa) = (50, 12), (36, 58/5). \quad (28)$$

For general transverse quartics the two κ combinations are $4\lambda_2 + 4\lambda_4 + 16\lambda'_4$ and $6\lambda_2 + 6\lambda_4 + 8\lambda'_4$ before multiplying by τ . The stationary ν^2 cancels all but $\tau\kappa\sigma^2$ in the mass, but *does not* cancel the fixed-Lagrangian mass derivative

$$b_\kappa = \nabla_r m_\kappa^2 = \left(\frac{12}{5}(\alpha - \beta)w, (\lambda_0 + 2\tau\kappa)\sigma, 120\chi_2 v_s \right). \quad (29)$$

This is the concrete reason that a light stationary spectrum can have large couplings to background fluctuations.

Proposition 2 (Conditional scalar kinetic lower bound). *Assuming these reducing-projector identities and a well-defined positive hard complement, the scalar hard kinetic matrix obeys*

$$(\Delta Z_S)_{ww} \geq \frac{(9/25)^2}{192\pi^2\tau\sigma^2} \left(\frac{50}{12} + \frac{36}{58/5} \right) = \frac{0.03107164552}{\tau}, \quad (30)$$

$$\rho_Z \geq \frac{(\Delta Z_S)_{ww}}{G_{ww}} \geq \frac{0.01294651897}{\tau}. \quad (31)$$

Proof. Each of the rank- N degenerate blocks contributes $N(9/25)^2 W(m^2, m^2)/(32\pi^2)$ to this diagonal kinetic entry. Since $W(m^2, m^2) = 1/(6m^2)$ and every omitted scalar-pair square has nonnegative weight, retaining the two blocks is a lower bound. The generalized Rayleigh quotient in the w direction supplies the second inequality. $\square$

Consequently $\tau < 0.0431551$ cannot satisfy the declared $\rho_Z < 0.3$ scalar gate on this ray under the stated identities. If another mode becomes tachyonic first, the candidate already fails its tree gate. The coefficient is obtained from the block masses and ranks, not a fit to the sampled ρ_Z values. This says nothing by itself about the complete gauge-dependent kinetic matrix or physical pole stability. It does show why continued weakening of these quartics alone is not a reliable repair strategy.

One can strengthen the bound without any further spectrum scan. Put $b_0 = (0.36, 0.11385, 0.03)$ and $z_0 = G^{-1}b_0/\sqrt{b_0^T G^{-1}b_0}$. Since all these components and τ are positive, $z_0^T b_\kappa \geq \sqrt{b_0^T G^{-1}b_0}$. Applying this single fixed witness to the sum of the two actual rank-one Gram blocks yields

$$\rho_Z \geq \frac{b_0^T G^{-1}b_0}{192\pi^2\tau\sigma^2} \left(\frac{50}{12} + \frac{36}{58/5} \right) = \frac{0.01471609503}{\tau}. \quad (32)$$

Thus $\tau < 0.0490537$ already fails the 0.3 scalar margin under the same verified-but-not-symbolically-certified action identities. This witness proof does not assume the generally false matrix inequality $b_\kappa b_\kappa^T \succeq b_0 b_0^T$.

6 Radial fermion kernel without a global family fit

The already checked Yukawa tensor map on this same radial family is $M_F = \sigma F_M$, with no w or v_s vertex. A Takagi transformation therefore simultaneously diagonalizes the radial mass and

its vertex: $M_i = \sigma f_i$, $y_i = \partial_\sigma M_i = f_i$, for three Majorana masses. The remaining family spinor directions have zero radial mass and vertex. The three f_i are inputs, not fitted numbers.

For one four-component Majorana fermion, expansion of $-\frac{1}{2} \text{Tr} \log(\not{\partial} + M + y\delta\sigma)$ gives

$$\Pi_F^{\text{raw}}(s) = 2y^2 \int_k \frac{M^2 - k \cdot (k+p)}{(k^2 + M^2)((k+p)^2 + M^2)} \quad (33)$$

$$= -\frac{y^2}{16\pi^2} [(4M^2 + s)L(M^2, M^2; s) + 2M^2(\ln(M^2/\mu_0^2) - 1)]. \quad (34)$$

The factor one half is Majorana counting; a Dirac field with the same couplings would contribute twice as much. The same determinant gives

$$V_F = -\frac{M^4}{32\pi^2} [\ln(M^2/\mu_0^2) - 3/2], \quad t_\sigma^F/\sigma = -\frac{2y^2 M^2}{16\pi^2} [\ln(M^2/\mu_0^2) - 1]. \quad (35)$$

Subtracting t_σ^F/σ by the same invariant finite mass counterterm, with no momentum-dependent tadpole subtraction, yields

$$\Pi_{\sigma\sigma}^{F,\text{FV}}(s) = -\frac{1}{16\pi^2} \sum_{i=1}^3 \frac{M_i^2}{\sigma^2} (4M_i^2 + s) L(M_i^2, M_i^2; s). \quad (36)$$

All other radial entries are zero. Its local finite kinetic term is

$$\Delta Z_{\sigma\sigma}^F = -\frac{1}{16\pi^2} \sum_{i=1}^3 \frac{M_i^2}{\sigma^2} [\ln(M_i^2/\mu_0^2) + 2/3]. \quad (37)$$

The canonical σ coefficient is half this expression because $G_{\sigma\sigma} = 2$. Both expressions have the continuous zero-Yukawa limit; no artificial $\log 0$ is evaluated. At zero momentum,

$$\Pi_{\sigma\sigma}^{F,\text{FV}}(0) \leq \frac{6\mu_0^4}{e 16\pi^2 \sigma^2} = 0.0875112373 \omega^2, \quad (38)$$

recovering P-REN1 by maximizing $-x^2 \ln(x/\mu_0^2)$ at $x = \mu_0^2 e^{-1/2}$. This bound is not asserted at nonzero momentum.

The independent verifier compares the closed form with the shifted spin-trace numerator integral

$$\Pi_F^{\text{raw}}(s) = \frac{2y^2}{16\pi^2} \int_0^1 D(x) [1 - 3 \ln(D(x)/\mu_0^2)] dx, \quad D(x) = M^2 + x(1-x)s, \quad (39)$$

followed by its tadpole subtraction. Five synthetic mass cards at five momenta, independent kinetic finite differences, zero-mass limits, the old curvature function and its exact cap pass 56/56 tests together with the gauge tensor checks below. These are regression inputs, not observed or fitted families. A full nonradial fermion kernel still requires the complex $h_{\text{raw}}, f_{\text{raw}}$ matrices.

7 What the gauge spectrum misses at nonzero momentum

To make the background-field label operational, choose the Euclidean quadratic gauge fixing, with real antisymmetric generators T^a ,

$$D_\mu X = \partial_\mu X + g a_\mu^a T^a X, \quad X = \bar{X} + \eta, \quad v_a = T^a \bar{X}, \quad F^a = \partial_\mu a_\mu^a + \xi g v_a^T \eta. \quad (40)$$

The quadratic bosonic operator and ghost operator are

$$Q_B = \begin{pmatrix} Q_{SS} & W^T \\ W & Q_{VV} \end{pmatrix}, \quad Q_{SS} = -\partial^2 + V'' + \xi g^2 \sum_a v_a v_a^T, \quad (41)$$

$$(Q_{VV})_{\mu\nu}^{ab} = [-\partial^2 \delta_{\mu\nu} + (1 - \xi^{-1}) \partial_\mu \partial_\nu] \delta^{ab} + M_{ab}^2 \delta_{\mu\nu}, \quad (42)$$

$$M_{ab}^2 = g^2 v_a^T v_b, \quad W_{\mu I}^a = -2g(T^a \partial_\mu \bar{X})_I, \quad Q_c = -\partial^2 + \xi M^2. \quad (43)$$

The mixed vertex follows directly from integrating the gauge-fixing cross term by parts. The scalar kinetic cross term is $ga_\mu[v_a^T\partial_\mu\eta - (\partial_\mu v_a)^T\eta]$; gauge fixing cancels the first term and doubles the second. Consequently $W = 0$ for a constant background, but not for a radial external momentum. With $d_{ar} = T^a R_r$, the actual additional tensors are

$$B_{r,ab} = g^2(d_{ar}^T v_b + v_a^T d_{br}), \quad (44)$$

$$C_{rs,ab} = g^2(d_{ar}^T d_{bs} + d_{as}^T d_{br}), \quad W_{r,\mu I}^a(p) = -2igp_\mu(d_{ar})_I. \quad (45)$$

Their numerical values and the identities $r_r B_r = 2M^2$, $r_r r_s C_{rs} = 2M^2$ are saved. The generalized eigenvalues of $g^2 \sum_a d_{ar}^T d_{as}$ are $(0, 3.1652201, 3.9565251)$. This is a vertex Gram matrix, *not* a loop kinetic correction.

The exact diagram generator is

$$\Gamma_1^B = \frac{1}{2} \text{Tr} \log Q_B - \text{Tr} \log Q_c + \Gamma_{\text{CT}}, \quad \Pi_{rs}^B = \frac{1}{2} \text{Tr}[Q_B^{-1} Q_{B,rs} - Q_B^{-1} Q_{B,r} Q_B^{-1} Q_{B,s}] + \Pi_{rs}^c + \Pi_{rs}^{\text{CT}}. \quad (46)$$

The remaining integrals include the two-vector bubble, vector seagull and scalar–vector derivative bubble. In strict Landau, $D_{a,\mu\nu}(k) = (\delta_{\mu\nu} - k_\mu k_\nu / k^2) / (k^2 + m_a^2)$, and the latter contains

$$\Pi_{rs}^{SV}(p) = -4g^2 \sum_{a,k} (d_{ar} \cdot u_k)(d_{as} \cdot u_k) \int_q \frac{p_\mu D_{a,\mu\nu}(q) p_\nu}{(q+p)^2 + m_k^2}. \quad (47)$$

Gauge generators must be rotated by the vector mass eigenbasis here. This dimensionally regulated integral has not yet been evaluated in the complete P54 prescription. Replacing the Lorentz trace by the integer three before pole subtraction loses finite $\epsilon \times 1/\epsilon$ terms. Similarly, direct background-scalar ghost vertices are proportional to ξ in this explicit fixing and can vanish in the strict Landau limit, but the finite- ξ /Nielsen check, and ghost terms in general matching, remain necessary. No guessed ghost counterterm is introduced.

8 Physical interpretation and outstanding work

Goldstone resummation of tadpoles alone does not make the second derivative of the Landau effective potential finite. External momentum is required; beyond one loop it must be combined with the appropriate resummation [1]. The cited two-loop mass construction specializes to the gaugeless limit and cannot be substituted wholesale for the present gauged theory. The VEV/tadpole convention must also be kept consistent with gauge dependence [2].

A physical stability claim requires the complete BRST-consistent inverse propagator and physical projection, including scalar–vector mixing when allowed, vector/longitudinal/ghost contributions, fermions, counterterms, analytic continuation and the relevant Ward/Nielsen identities. Neither a negative partial hard curvature nor an isolated root of a scalar-only kernel is such a claim. Full Wilson/box matching, lower gauge/ghost/Yukawa finite matching and the complete finite seesaw including C_{HN} remain open. A flavor fit is not promoted.

References

- [1] J. Braathen and M. D. Goodsell, “Avoiding the Goldstone Boson Catastrophe in general renormalisable field theories at two loops,” [arXiv:1609.06977](#).
- [2] V. Dūdėnas and M. Löschner, “Vacuum expectation value renormalization in the Standard Model and beyond,” [arXiv:2010.15076](#).